

```
from google.colab import files
uploaded = files.upload()
```

Screenshot ... 142020.png

Screenshot 2026-07-13 142020.png(image/png) - 268920 bytes, last modified: 13/07/2026 - 100% done
Saving Screenshot 2026-07-13 142020.png to Screenshot 2026-07-13 142020 (3).png

```
import cv2
import numpy as np
import matplotlib.pyplot as plt

filename = list(uploaded.keys())[0]
img = cv2.imread(filename, 0)

# Laplacian mask with diagonal neighbors
mask = np.array([
    [-1, -1, -1],
    [-1,  8, -1],
    [-1, -1, -1]
], dtype=np.float32)

lap = cv2.filter2D(img, cv2.CV_32F, mask)

# Sharpen image
sharp = cv2.addWeighted(img.astype(np.float32), 1, lap, 1, 0)
sharp = np.clip(sharp, 0, 255).astype(np.uint8)

plt.imshow(sharp, cmap='gray')
plt.title('Laplacian Sharpened Image')
plt.axis('off')
plt.show()
```

Laplacian Sharpened Image

Successfully downloaded input.jpg



Cause of Aliasing:

- Aliasing occurs when the image is sampled below the Nyquist rate.
- High-frequency details appear as jagged edges or moiré patterns.
- Quantization reduces pixel intensity precision but is not the main cause of

Corrective Approach:

- Apply an anti-aliasing (Gaussian) filter before downsampling.
- Use better interpolation methods such as INTER_LINEAR or INTER_AREA.
- Increase the sampling resolution if possible